

DK-AOS

(Alginate oligosaccharide)

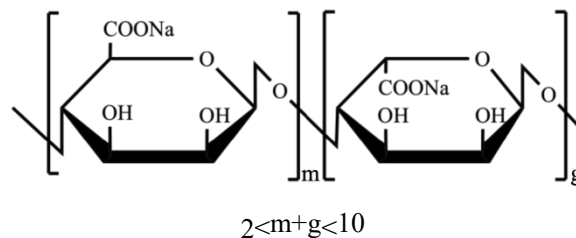
New Bioactive Compound From Brown Algae

【Company Introduction】

Qingdao DaKang Marine Biotechnology Co., Ltd. is located in Qingdao West Coast New Area. It is a marine oligosaccharide industrialization base jointly built by the Qingdao Marine Biomedical Research Institute and Qingdao Gather Great Ocean Algae Industry Group Co., Ltd. The expert team is spearheaded by Professor Guan Huashi, a distinguished modern marine pharmaceutical scientist and academician of the Chinese Academy of Engineering, together with his research group. The company has always focused on the research, development, and production of marine functional oligosaccharides, establishing a series of industrial chain production bases for products such as alginate oligosaccharides, carrageenan oligosaccharides, agar oligosaccharides, and high-quality sodium alginate. The production base covers an area of approximately 2,000 square meters, equipped with a Class C clean workshop and advanced industry facilities, enabling large-scale and standardized production of high-quality marine functional oligosaccharides. This promotes comprehensive applications in fields such as healthcare, daily chemicals, medical devices, and more, fostering the emergence of a billion-yuan-level industrial cluster.

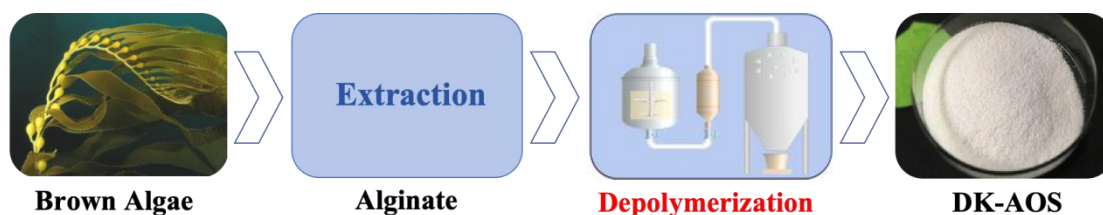
【Brief Introduction】

DK-AOS is composed of D-mannuronic acid and L-glucuronic acid with a degree of polymerization (DP) ranging from 2 to 10.



【Manufacturing Process】

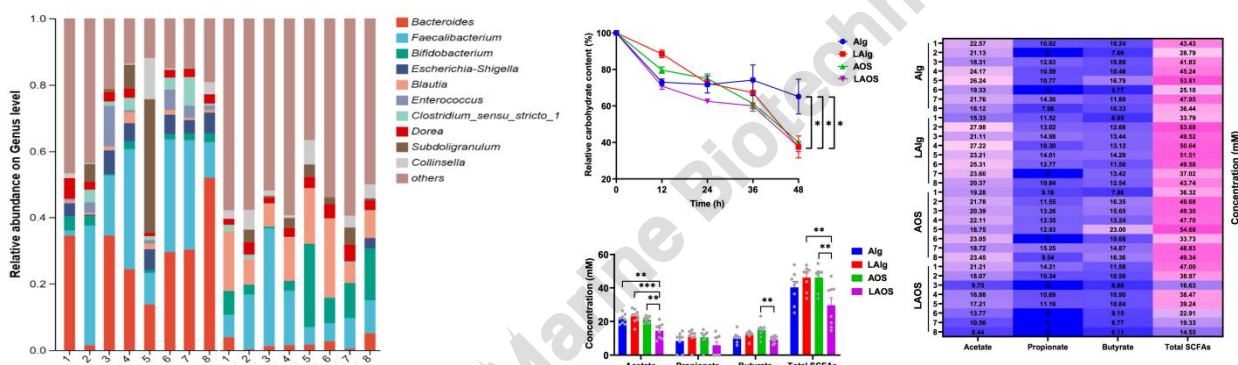
DK-AOS is obtained from brown algae through extraction, hydrolysis, purification, and drying.



【Biological Activities】

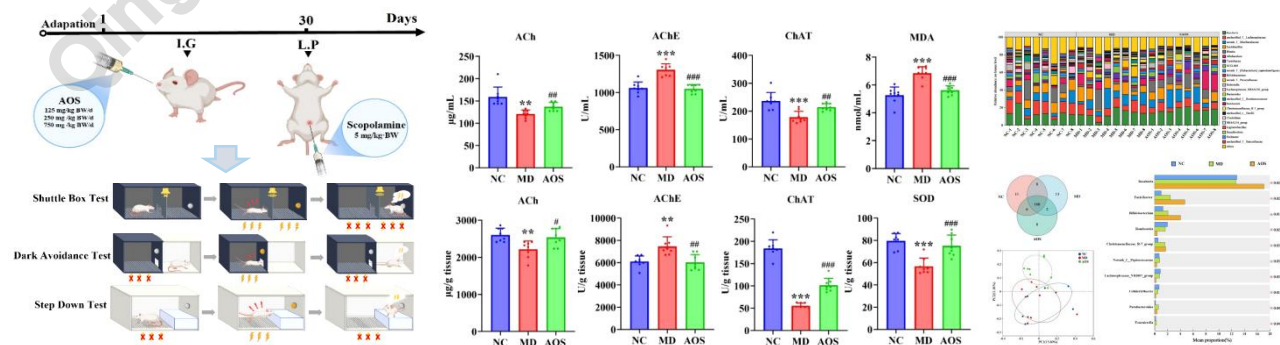
1) Marine functional factor

DK-AOS is a new marine functional factor which can significantly promote the abundance of new-generation probiotics such as *Bacteroides xylanisolvens* and *Faecalibacterium prausnitzii*, facilitate the synthesis of short-chain fatty acids, and improve the intestinal immunity.



2) Improving memory

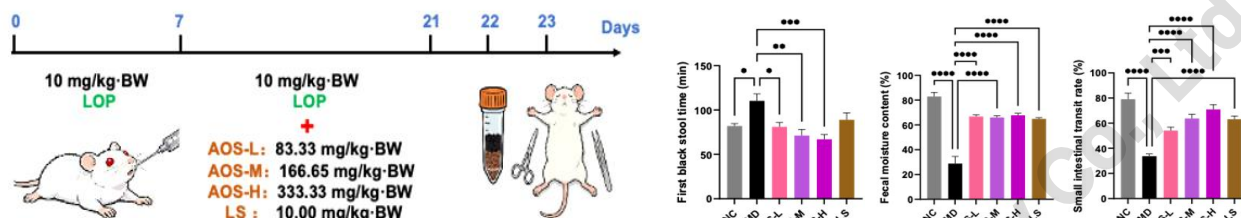
It has been verified by animal experiments that DK-AOS has a significant effect on improving memory in normal aging animals and scopolamine-induced memory impairment animals. It can ameliorate the scopolamine-induced reduction in *Roseburia* abundance and interfere with the endogenous cannabinoid system and cholinergic system, and play a cognitive improvement role through a synergistic network.



3) loosening bowel to relieve constipation

It has been verified by animal experiments that DK-AOS can significantly shorten the first

defecation time of experimental animals, improve the small intestine propulsion rate, increase the water content of feces, and restore the integrity of intestinal wall villi, which has a significant laxative effect.



【Specification】

Analytical Test	Specification
● Apperance	Pale yellow or yellow powder, odorless, with no visible foreign impurities to the naked eye.
● Content (%)	≥90.0
● Average Molecular Weight (Da)	600~1400
● Average Degree of Polymerization	3~7
● Loss on Drying (%)	≤10.0
● Ash (%)	20.0~27.0
● pH (1%, Solution)	5.5~6.5
● Pb (mg/kg)	≤2.0
● As (mg/kg)	≤1.0
● Hg (mg/kg)	≤0.3
● Total Plate Count (CFU/g)	≤1000
● Coliform bacteria (CFU/g)	≤10
● Mold and Yeast (CFU/g)	≤50
● Staphylococcus Aureus (/25g)	Not detectable
● Salmonella (/25g)	Not detectable

【Applications】

- Functional Foods
- Nutraceuticals
- Dietary Supplements
- Medicinal Foods



【Manufacturer】

Infomation	Contents
● Name	Qingdao Dakang Marine Biotechnology Co., Ltd
● Mission	To be the manufacturer of marine bioactive oligosaccharides
● Main Field of Business	Seaweed Oligosaccharides (Alginate oligosaccharide, DK-AOS; Agaro oligosaccharide, DK-AGOS; Carrageenan oligosaccharide, DK-CAOS)
● Email	DK@judayang.com